

Week 4 – Self-Study Tasks

Understanding the Transactions

1. What does one transaction represent?

One transaction represents one case where a set of items or events happened together, for example one student-week or one quiz attempt.

2. What are the possible items?

Items are the events or conditions recorded in a transaction, such as quiz completed, task submitted or activity level.

3. Which columns are identifiers rather than items?

Columns such as student ID or transaction ID are identifiers. They identify the record but should not be treated as items.

4. Are any items duplicates or inconsistent?

This should be checked before the analysis because duplicates or inconsistent names could affect the results.

5. Are any values missing?

Missing values should be checked first. I would not automatically treat a missing value as an event not happening.

6. Are all proposed items meaningful for the question?

Not always. I would only use items that have a clear meaning for what I am trying to investigate.

Transaction Preparation

7. What does True in one item column mean?

True means that the item is present in that transaction.

Frequent Itemsets

8. What happens to the number of frequent itemsets when minimum support is lowered?

More itemsets are found because less common combinations can now pass the minimum support threshold.

Threshold Sensitivity

9. What happens when support is lowered?

More itemsets and rules usually appear, including some rare ones.

10. Do very rare rules appear?

Yes, especially when the support threshold is low. I would be careful about trusting them.

11. What happens when confidence is increased?

Fewer rules remain because the requirement becomes stricter.

12. Which rules disappear when lift greater than 1 is required?

Rules with lift of 1 or below disappear because they do not show a positive association.

13. Can a rule be statistically interesting but practically unimportant?

Yes. A rule can have good numbers but occur so rarely that it is not very useful in practice.

Interpretation, Validity and Ethics

14. Are there enough transactions to trust rare patterns?

Rare patterns need to be treated carefully. I would want more data before making a strong conclusion.

15. Are repeated transactions from the same synthetic student independent observations?

Not completely, because several transactions can come from the same student and may be related.

16. Could the way items were constructed create the association?

Yes. How we create categories or choose thresholds can affect which associations appear.

17. Could an association be explained by a third variable?

Yes. Another variable could be affecting both items, so association does not mean causation.

18. Does the dataset observe the behaviour you think it observes?

Not completely. It only shows what the system records, so missing activity does not necessarily mean no study activity happened.

19. Could a rule lead to an unfair label about a student or group?

Yes. The rule only describes patterns in the data. It should not be used to label someone as a certain type of student.

20. Could a small combination of items make individuals identifiable?

Yes, especially if the combination is rare and the dataset is small.

21. What evidence would be needed before using a rule in a real decision?

I would need more data, consistent results and a clear reason why the rule matters. I would also check other possible explanations and privacy or fairness issues.